

Chapter 1: Introduction to Classification

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1 Introduction to Machine Learning

1.1 Objectives

By the end of this section, you will be able to:

- Define Machine Learning.
- Explain why Machine Learning is important.
- Introduce the three main learning paradigms.

1.2 Definition of Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.

1.3 Artificial Intelligence vs. Machine Learning

While Artificial Intelligence is the broad science of mimicking human abilities, Machine Learning is the specific suite of algorithms and methods that train a model on data to make predictions or decisions.

1.4 Why Machine Learning?

Traditional programming requires human developers to write explicit rules. However, for complex tasks like image recognition or natural language understanding, writing manual rules is nearly impossible. Machine Learning allows computers to discover these underlying patterns automatically from raw data.

1.5 Real-life Examples

- **Email Filters:** Automatically routing spam away from your inbox.
- **Ride-Sharing Apps:** Predicting arrival times and dynamic pricing.

- **Voice Assistants:** Translating spoken words into text commands (e.g., Siri, Alexa).

1.6 Types of Machine Learning

Supervised Learning: The model is trained on labeled data, meaning the inputs are paired with correct answers.

Unsupervised Learning: The model analyzes unlabeled data to find hidden patterns, structures, or groupings.

Reinforcement Learning: An agent learns to make decisions by performing actions in an environment to maximize a reward.

2 Supervised Learning

Since classification is a subset of supervised learning, it is essential to understand the basic mechanics of this learning paradigm.

2.1 Key Concepts

Input Features (X): The independent variables or attributes used to make a prediction.

Target Variable (y): The dependent variable or label we want the model to predict.

Training Dataset: The historical data used to help the model learn the relationship between X and y .

Testing Dataset: An unseen portion of the dataset used to evaluate the model's true performance.

2.2 Illustrative Example

Consider the following customer dataset:

Age	Salary (\$)	Purchased?
25	35,000	Yes
41	80,000	Yes
19	15,000	No

Table 1: Sample Customer Data

In this dataset:

- Input Features (X) = Age, Salary
- Target Variable (y) = Purchased (Yes/No)

3 What is Classification?

3.1 Definition & Objective

Classification is a supervised learning task where the output variable is a discrete category (or class). The primary objective is to build a mapping function from input features (X) to identify the correct category (y).

3.2 How a Classifier Learns

During training, the algorithm looks at thousands of examples to find patterns. It seeks to draw a **decision boundary**—a conceptual line or surface that separates different classes in the feature space. Once this boundary is established, new, unseen data points are categorized based on which side of the boundary they fall.

3.3 Core Examples

- **Spam Detection:** Labeling an email as *Spam* or *Not Spam*.
- **Healthcare:** Diagnosing a tumor as *Cancerous* or *Healthy*.
- **Finance:** Flagging a transaction as *Fraudulent* or *Legitimate*.
- **Computer Vision:** Identifying an image as a *Cat* or a *Dog*.

4 Classification vs. Regression

It is easy to confuse classification with regression, but they serve fundamentally different tasks:

Regression	Classification
Predicts continuous numbers	Predicts discrete categories
Example: House Price Prediction	Example: Spam Detection
Example: Temperature Forecasting	Example: Disease Diagnosis
Continuous Output	Discrete Output

Table 2: Comparison of Regression and Classification

5 Types of Classification

Classification tasks can be broken down based on the structure of their target variables:

Binary Classification: Predicting one of exactly two possible outcomes.

- *Examples:* Yes / No, Pass / Fail, Disease / Healthy.

Multi-class Classification: Predicting one category out of three or more mutually exclusive outcomes.

- *Examples:* Handwritten Digits (0–9), Iris Species (Setosa, Versicolor, Virginica), Animal Recognition (Cat, Dog, Bird).

Multi-label Classification: Assigning multiple categories/labels to a single data point simultaneously.

- *Example (Movie):* A film can be labeled as both **Action**, **Adventure**, and **Comedy**.
- *Example (Image):* An image can contain labels like **Dog**, **Pet**, **Outdoor**, and **Animal**.

6 Real-World Applications

Healthcare: Used in disease risk prediction and automating cancer diagnoses from medical imaging.

Finance: Powering real-time credit card fraud detection and determining risk scores for loan and credit approvals.

Cybersecurity: Detecting malware variants and identifying network intrusion patterns.

Social Media: Running sentiment analysis to detect positive, negative, or neutral user opinions.

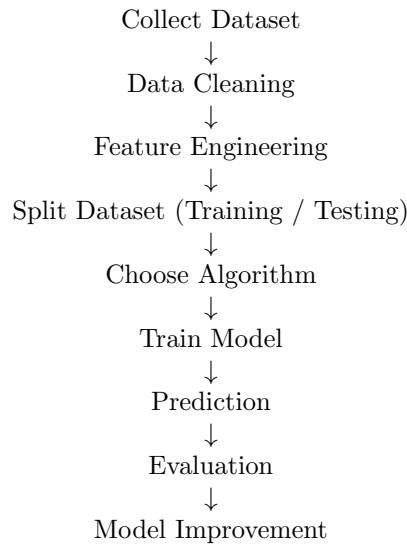
Computer Vision: Powering facial recognition systems on smartphones and object detection in autonomous vehicles.

Natural Language Processing: Auto-filtering spam emails and detecting the language of written text.

Recommendation Systems: Suggesting product categories or movies to users based on prior purchases.

7 Classification Workflow

Building a successful classification model requires a highly structured, iterative pipeline:



1. **Collect Dataset:** Gathering relevant historical data.
2. **Data Cleaning:** Handling missing values, removing duplicates, and dealing with outliers.
3. **Feature Engineering:** Transforming raw variables into patterns that algorithms can easily process.
4. **Split Dataset:** Dividing the dataset (usually 80/20) into a training set to build the model and a testing set to evaluate it.
5. **Choose Algorithm:** Selecting the model type best suited for the nature of the data.
6. **Train Model:** Letting the algorithm learn patterns from the training dataset.
7. **Prediction:** Running the trained model on unseen test data to generate labels.
8. **Evaluation:** Using testing metrics to measure how well the model predicted the correct labels.
9. **Model Improvement:** Fine-tuning parameters or updating features to maximize accuracy and minimize error.

8 Evaluation Metrics

To assess whether a model works well, we rely on concrete evaluation metrics.

8.1 Confusion Matrix

The cornerstone of evaluation is the **Confusion Matrix**, a tabular summary of predicted versus actual labels for a binary classifier.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Table 3: Generic Confusion Matrix

- **True Positive (TP):** Model correctly predicted a positive outcome.
- **False Negative (FN):** Actual state was positive, but the model incorrectly predicted negative (Type II error).
- **False Positive (FP):** Actual state was negative, but the model incorrectly predicted positive (Type I error).
- **True Negative (TN):** Model correctly predicted a negative outcome.

8.2 Numerical Example

Suppose we test our model on **100 patients** where:

- Actual Positive = 50, Actual Negative = 50.
- The model outputs: TP = 42, FN = 8, FP = 5, TN = 45.

	Predicted Positive	Predicted Negative
Actual Positive	42	8
Actual Negative	5	45

Table 4: Example Confusion Matrix

8.3 Metric Formulas & Calculations

8.3.1 Accuracy

Accuracy represents the percentage of total predictions that were correct.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Using our numerical example:

$$\text{Accuracy} = \frac{42 + 45}{100} = 0.87 \quad (87\%)$$

8.3.2 Precision

Precision tells us what fraction of the predicted positive cases were actually positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Using our numerical example:

$$\text{Precision} = \frac{42}{42 + 5} \approx 0.894 \quad (89.4\%)$$

Meaning: Among all cases the model flagged as positive, 89.4% were true positives.

8.3.3 Recall (Sensitivity)

Recall measures the percentage of actual positive cases that our model successfully flagged.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Using our numerical example:

$$\text{Recall} = \frac{42}{42 + 8} = 0.84 \quad (84.0\%)$$

Meaning: The model successfully detected 84% of all real positive cases.

8.3.4 F1-Score

The F1-Score is the harmonic mean of Precision and Recall. It is highly useful when looking for a balance between the two metrics.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Using our numerical example:

$$\text{F1-Score} = 2 \cdot \frac{0.894 \times 0.84}{0.894 + 0.84} \approx 0.866 \quad (86.6\%)$$

8.4 Which Metric Should We Use?

Different real-world scenarios dictate which evaluation metric to prioritize:

Situation	Best Metric	Reasoning
Balanced Dataset	Accuracy	Overall performance is descriptive and reliable.
Spam Detection	Precision	Avoid marking legitimate, critical emails as spam.
Cancer Detection	Recall	Missing a sick patient is far costlier than a false alarm.
Imbalanced Dataset	F1-score	Balances Precision and Recall where data is lopsided.

Table 5: Selecting the Right Classification Metric

9 Classification Algorithms Overview

Below is a brief look at the seven primary algorithms used to handle classification tasks:

Order	Algorithm	Main Idea
1	Logistic Regression	Predicts class probabilities using the sigmoid function.
2	Decision Tree	Makes predictions by following hierarchical decision rules.
3	K-Nearest Neighbors	Predicts a sample's label based on the class labels of its nearest neighbors.
4	Naive Bayes	Applies Bayes' theorem with strict feature independence assumptions.
5	Support Vector Machine	Finds the optimal separating hyperplane that maximizes the margin between classes.
6	Random Forest	Combines multiple decision trees together through bagging to improve accuracy.
7	Gradient Boosting Machine	Sequentially builds decision trees where each new tree corrects previous errors.

Table 6: Summary of Common Classification Algorithms